

Common Nasal Conditions & Nasopharyngeal Carcinoma

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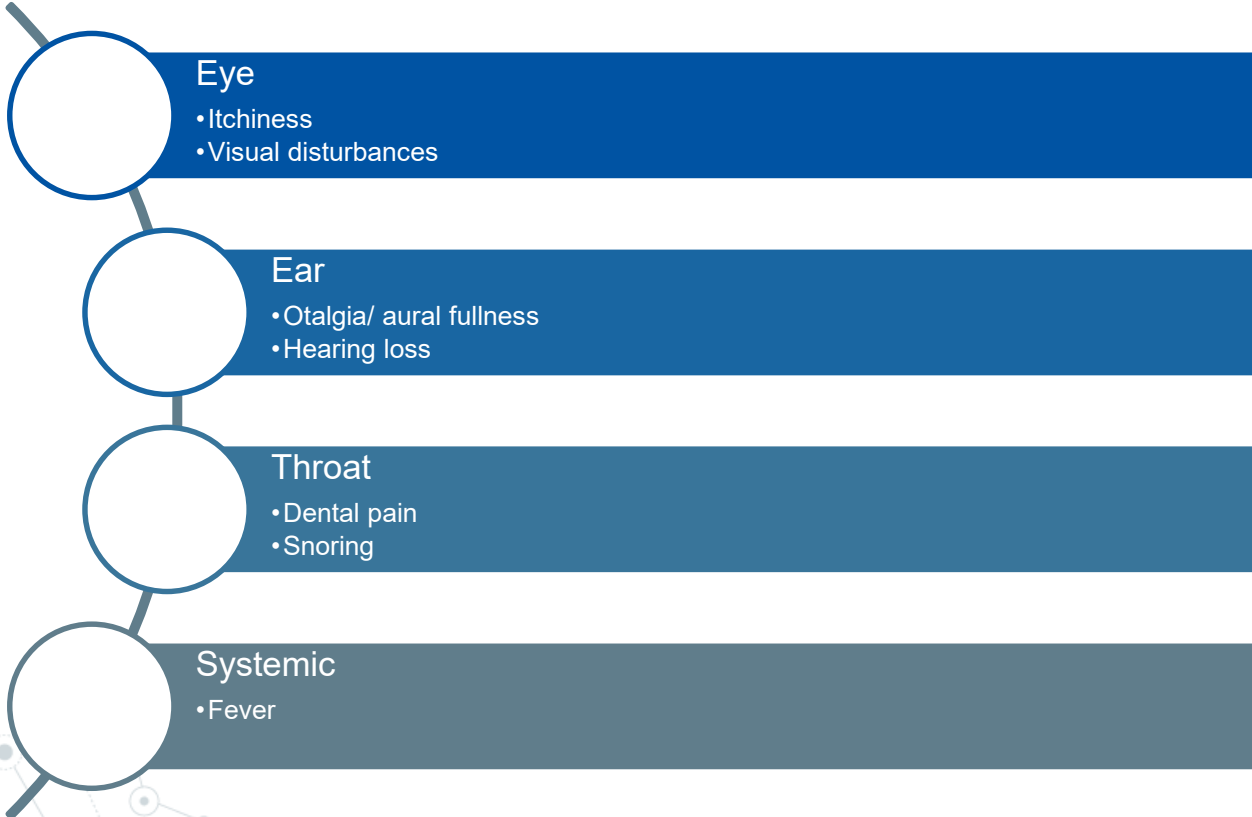
Learning Objectives

- ◎ Understand the pathophysiology of common nasal conditions
- ◎ Medical and surgical management of common nasal conditions
- ◎ Clinical presentation of nasopharyngeal carcinoma
- ◎ Management of nasopharyngeal carcinoma

History Taking

- ◎ Congestion/ blockage/ obstruction
- ◎ Nasal discharge
 - Anterior/ posterior
 - Nature
- ◎ Sneezing, itchiness
- ◎ Olfactory disturbances/ loss of smell
 - Loss of taste ❌
- ◎ Facial pressure/ pain
- ◎ Epistaxis

History Taking



History Taking

- ◎ Other comorbidities: eg asthma, atopy, immunodeficiency
- ◎ Social and occupational history
- ◎ Drug history
- ◎ Smoking
- ◎ Family history

Physical Examination

◎ External nose



Image Source: Logan Turner Disease of the Nose, Throat and Ear- Head and Neck Surgery

Physical Examination

🎯 Anterior rhinoscopy



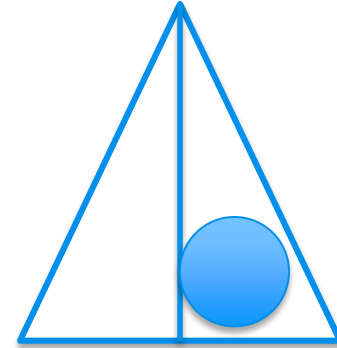
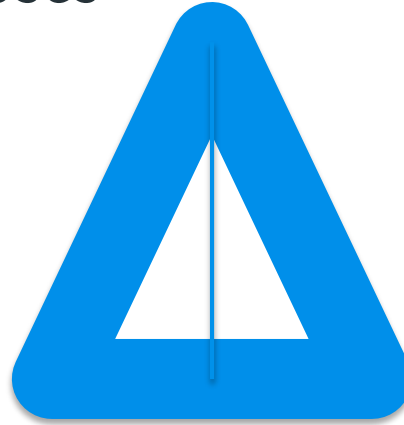
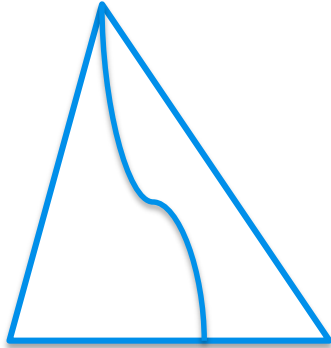
Nasoendoscopy

- ◎ Rigid
- ◎ Flexible



Nasal Obstruction

◎ Differential Diagnoses



Nasal deformity

- ◎ Crooked nose
- ◎ Deviated nose
- ◎ Saddle nose

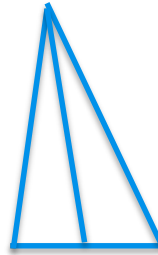
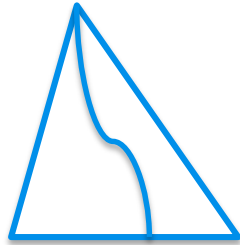


Image source: Disease of Ear, Nose and Throat

Injuries of the Nose

◎ Septal haematoma

- Collection of blood under perichondrium
- Risk of ischaemia, septal perforation and collapse
- Infection → meningitis, cavernous sinus thrombosis
- Urgent incision and drainage

◎ Fracture nasal bone

◎ Fracture or dislocation of the septum

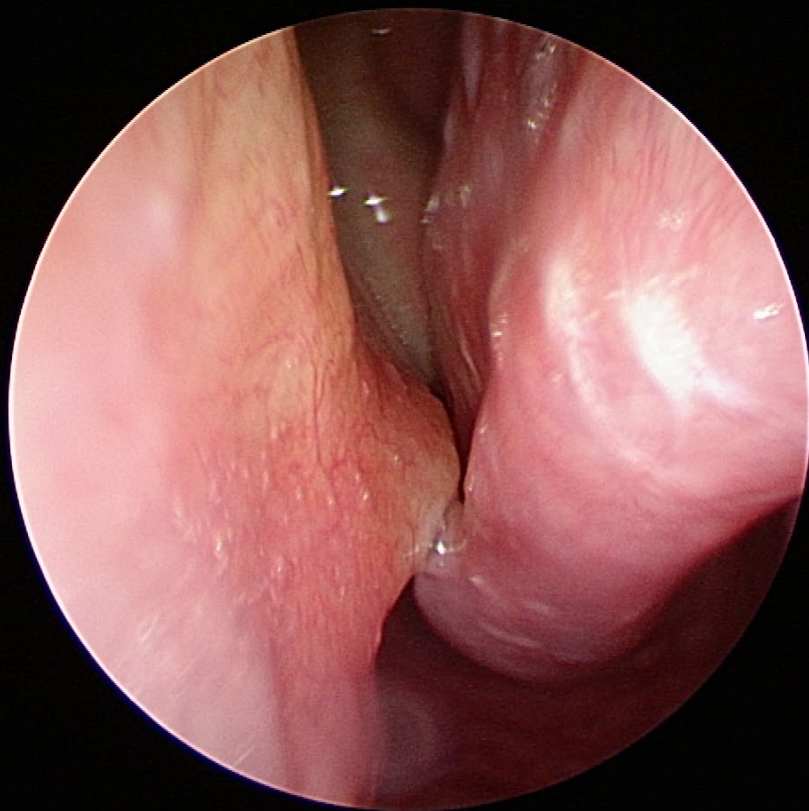
Injuries of the Nose

◎ Fracture nasal bone

- May have open wound in the skin
- Swelling/ discoloration of skin over nasal bone
- Tenderness
- Mobility of the nose
- Deformity
- Treatment
 - ◎ Treat epistaxis, open wound
 - ◎ No treatment if no deformity
 - ◎ Closed reduction within 7-10 days

Nasal Septal Deviation

- ◎ Rarely exactly in midline
- ◎ Can be asymptomatic if minor
- ◎ Causes
 - Trauma (birth, long-forgotten or recent)
 - Nasal surgery
- ◎ Marked deviation
 - Nasal obstruction
 - Obstruction to normal sinus drainage pathway
 - Epistaxis
- ◎ Management
 - Septoplasty or septorhinoplasty if symptomatic



Mucosal Swelling

- ◎ Acute infection
- ◎ Chronic rhinosinusitis
 - With or without polyposis
- ◎ Allergic rhinitis
- ◎ Non-allergic rhinitis



Acute Infections

- ◎ Upper respiratory infection
- ◎ Nasal vestibulitis
 - Skin of nasal vestibule
 - Staphylococci
 - Topical antibiotics ointment

Rhinosinusitis

- ◎ Paranasal sinuses lined with nasal mucosa
- ◎ Subject to same diseases as the nose
- ◎ 'Rhinosinusitis' more accurate than 'sinusitis'

Acute Bacterial Rhinosinusitis

◎ Causes

- Upper respiratory tract infection
- Dental abscess/ extraction (pre-molar/ molar teeth)
- Trauma

◎ Predisposing factors

- Poor drainage e.g. septal deviation, turbinate hypertrophy, nasal polyposis
- Poor immunity

Acute Bacterial Rhinosinusitis

- ⊙ Symptoms of viral URTI > 10 days or worsening after 5-7 days (double sickening)
- ⊙ Nasal obstruction
- ⊙ Nasal discharge (anterior or post-nasal drip)
- ⊙ Facial pain
- ⊙ Reduction of smell
- ⊙ Fever

Acute Bacterial Rhinosinusitis

A decorative network diagram in the top right corner, consisting of a complex web of interconnected nodes and lines, resembling a molecular or biological structure.

◎ Organisms

- *Streptococcus pneumoniae*
 - *Haemophilus influenzae*
 - *Moraxella catarrhalis*
 - Anaerobic organisms (dental source)
- 
- A decorative network diagram in the bottom left corner, consisting of a complex web of interconnected nodes and lines, resembling a molecular or biological structure.

Acute Bacterial Rhinosinusitis

⊙ Signs

- Facial tenderness
- Edema, mucopurulent discharge in middle meatus/nasopharynx

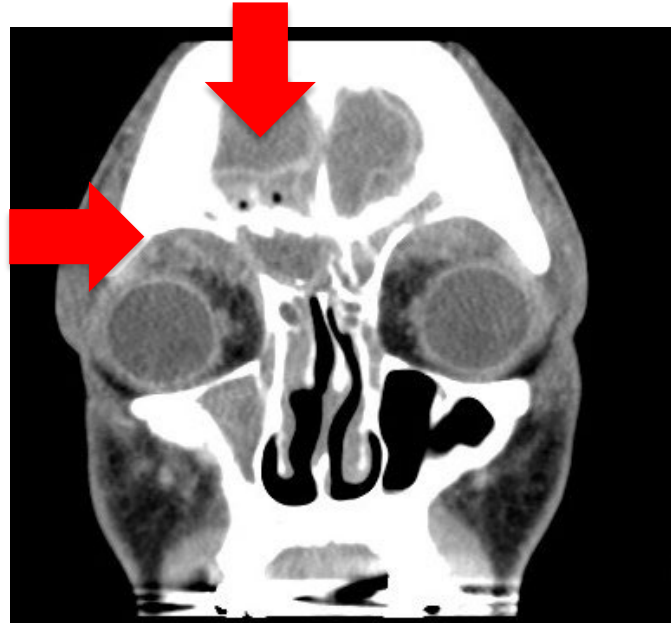
⊙ Treatment

- Analgesics
- Antibiotics
- Intranasal steroid spray
- Short-term (<7 days) nasal decongestant e.g. oxymetazoline, ephedrine
- Nasal douching

Acute Bacterial Rhinosinusitis

⊙ Complications

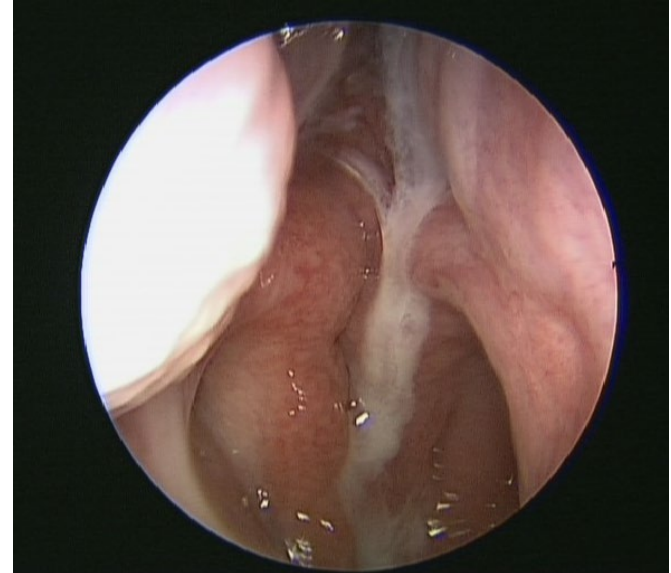
- Orbital cellulitis, abscess
- Cavernous sinus thrombosis
- Intracranial infection
 - ⊙ Meningitis, encephalitis, abscess



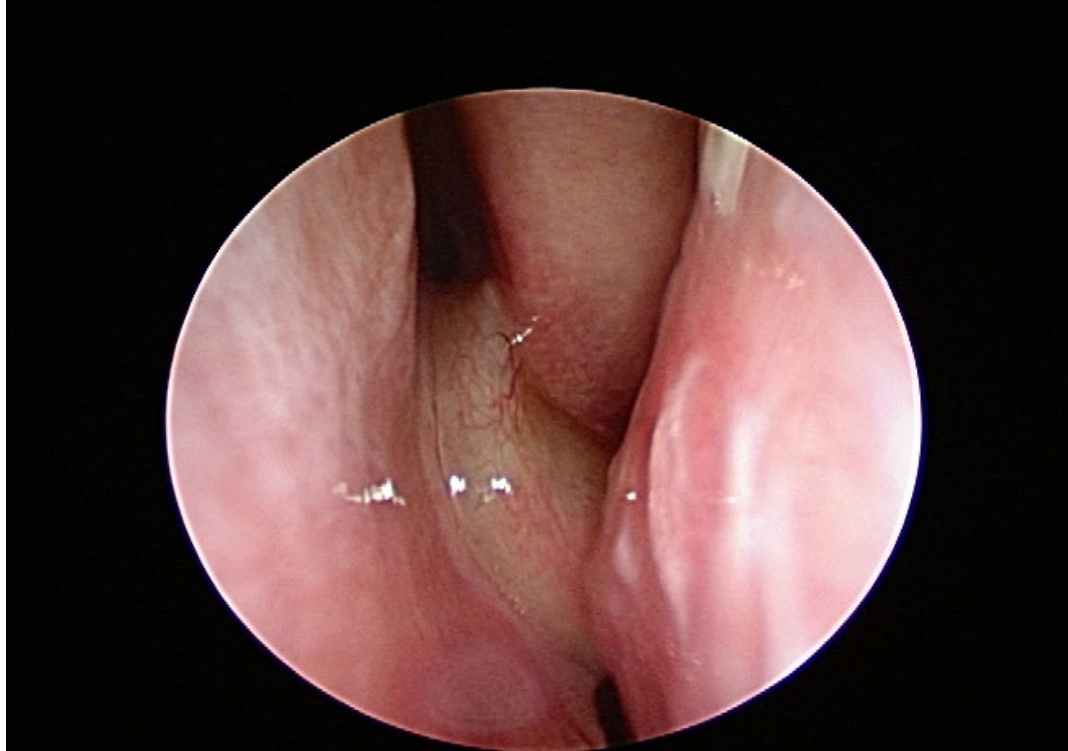
Chronic Rhinosinusitis

© Clinical features:

- Purulent nasal and post-nasal discharge
- Nasal obstruction
- Facial discomfort
- Headache
- Halitosis



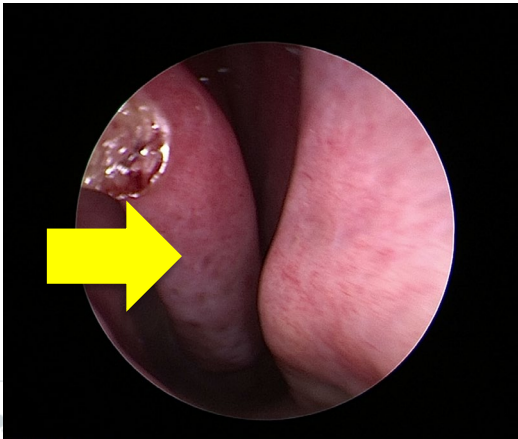
Chronic Rhinosinusitis with polyposis



Nasal polyposis

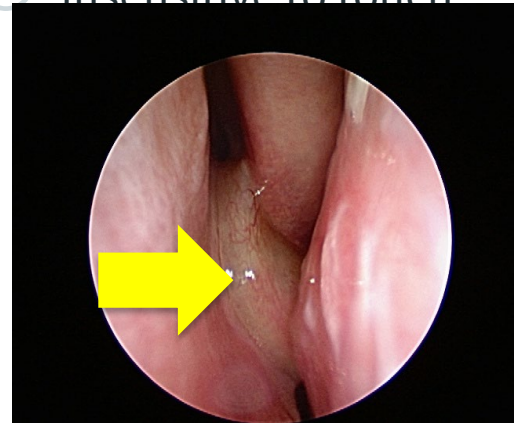
Inferior Turbinate

- Sensitive
- Attached to lateral nasal wall



Nasal polyp

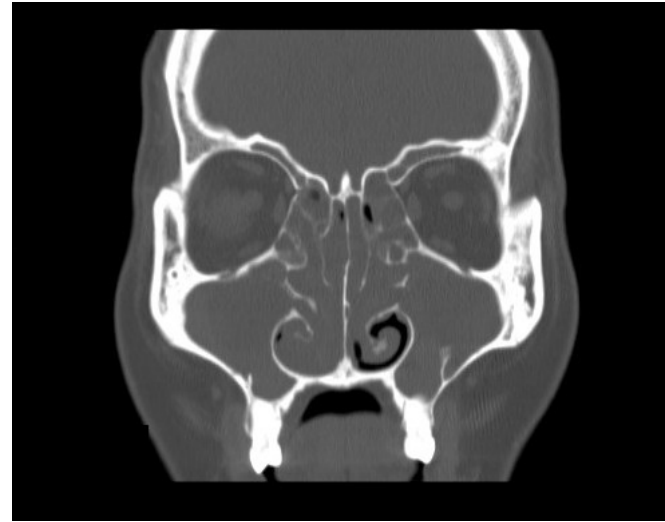
- Pale, greyish
- Translucent
- Insensitive to touch



Nasal polyposis

Associated diseases

- Asthma
 - Samter's triad
- Allergic fungal sinusitis
- Cystic fibrosis



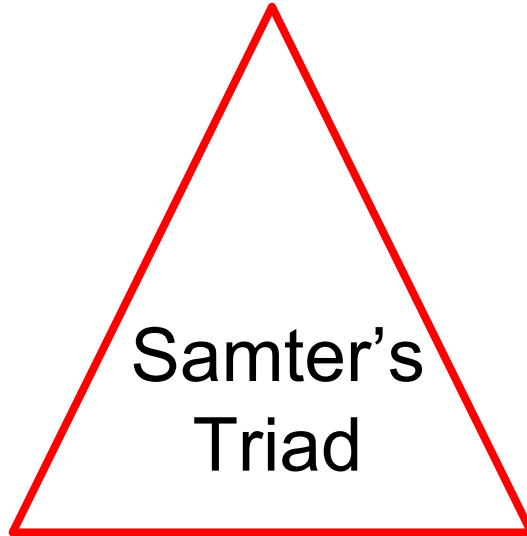
Aspirin Exacerbated Respiratory Disease

Aspirin
Sensitivity

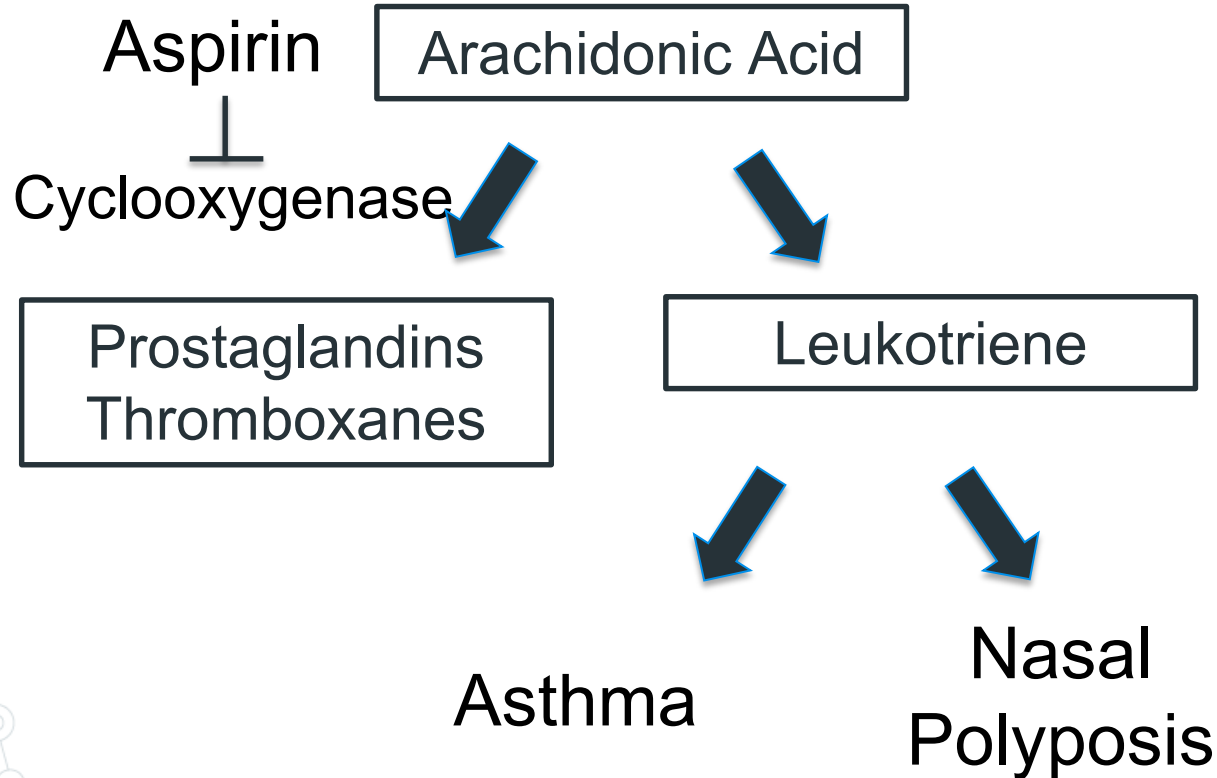
Samter's
Triad

Asthma

Nasal
Polyposis



Aspirin Exacerbated Respiratory Disease

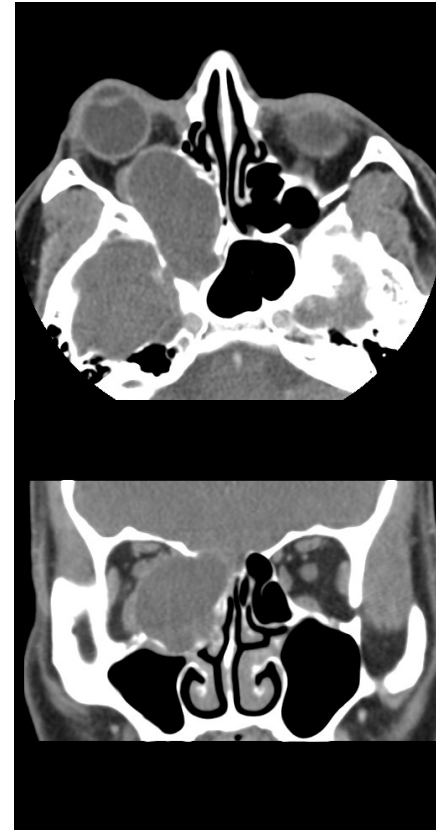


Chronic Rhinosinusitis

- ◎ **Intranasal steroid**
- ◎ Short-term antibiotics for superimposed infection
- ◎ Long-term antibiotics
 - Anti-inflammatory effect
 - Macrolide, doxycycline
- ◎ **Nasal saline irrigation**
- ◎ Anti-histamine for atopy
- ◎ Surgery: endoscopic sinus surgery

Mucoceles

- ◎ Epithelial-lined sac
- ◎ Contains mucus
- ◎ Expand and erode bone
- ◎ Most common in the fronto-ethmoidal region
- ◎ Orbital displacement and proptosis
- ◎ Surgical marsupialization



Allergic Rhinitis

- ◎ Type I hypersensitivity reaction
- ◎ Aeroallergens – Immunoglobulin E
- ◎ United airway disease
 - Common epidemiology, pathophysiology, treatment
 - AR patients: 15-38% have asthma
 - Asthma patients: 6-85% have nasal symptoms
 - Improve asthma control with treatment of allergic rhinitis

Allergic Rhinitis

⊙ Symptoms

○ Early

- ⊙ Sneezing
- ⊙ Itching
- ⊙ Rhinorrhoea
- ⊙ Nasal obstruction

○ Late

- ⊙ Nasal congestion
- ⊙ Hyperresponsiveness
 - Allergens
 - Irritants, atmospheric changes

Allergic Rhinitis

History

- Other atopy e.g. asthma, eczema, allergic conjunctivitis
- Possible triggers
 - Season
 - Pets
- Intermittent vs persistent symptoms, severity
- Associated problems
 - Sinusitis
 - Otitis media
 - Sleep disturbance

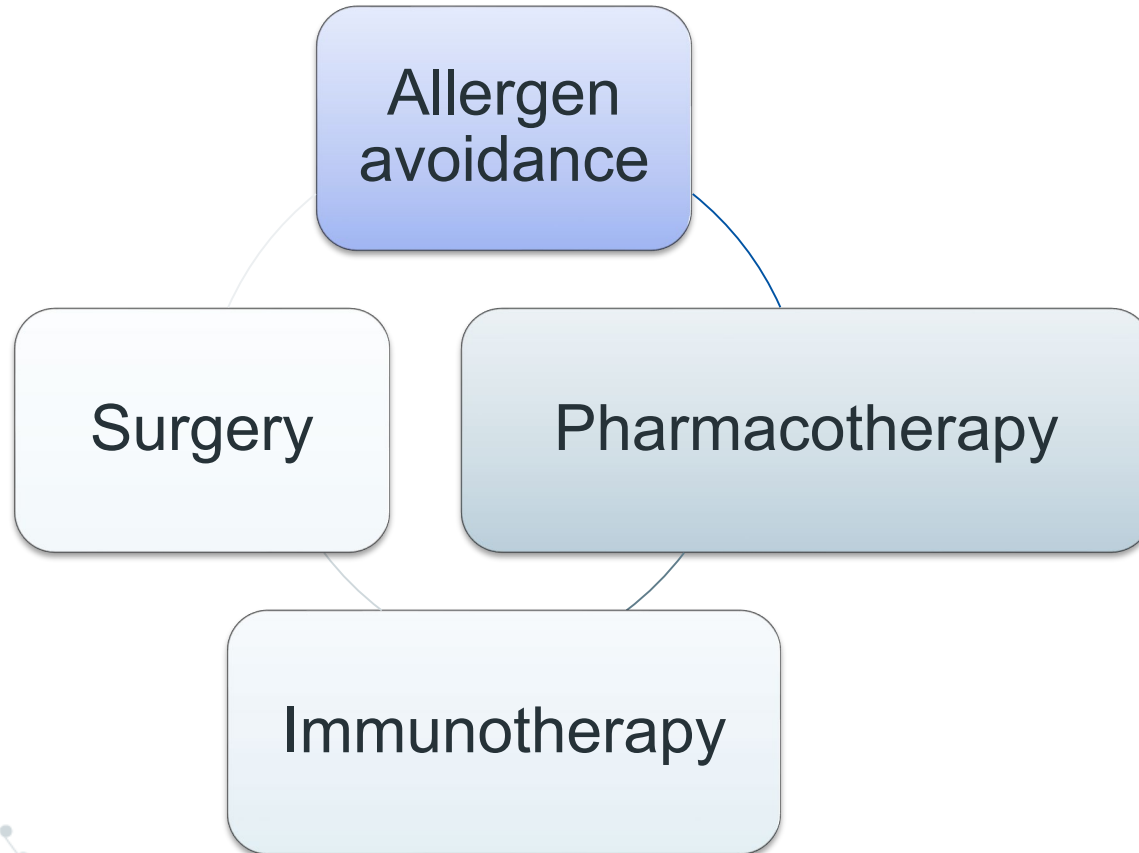
Allergic Rhinitis

◎ Common allergen in Hong Kong

- House dust mites
- Cockroach
- Furry Pet
- Pollen
- Mold
- Multi-allergen



Allergic Rhinitis



Allergic Rhinitis

◎ Pharmacotherapy

- Oral antihistamine
- Intranasal steroid
- Leukotriene receptor antagonists
 - ◎ Especially in patients with asthma
- Saline irrigation
- Refractory: short term systemic steroid, decongestant

Allergic Rhinitis

- ◎ Skin Prick Test
- ◎ In vitro specific IgE blood test
 - Radioallergosorbent test (RAST)
- ◎ Immunotherapy
 - Subcutaneous
 - Sublingual



Non-allergic Rhinitis

- Idiopathic – temperature, humidity, pressure (vasomotor rhinitis, intrinsic)
- Drug-induced
- Food-induced
- Hormonal
- Irritants
- Occupational
- NARES – non-allergic rhinitis with eosinophilia syndrome
- Atrophic rhinitis
- Emotional
- Gastro-esophageal reflux disease
- Autonomic

Drug-induced Rhinitis

- ◎ Anti-hypertensive
 - β -blockers
 - Calcium channel blockers
- ◎ Sedatives
- ◎ Antidepressants
- ◎ Oral contraceptives
- ◎ Etc

Drug-induced Rhinitis

⊙ Rhinitis medicamentosa

- Reactive vasodilatation of nasal mucosa
- Acquired sensitivity of nasal lining after prolonged use of topic sympathomimetic agents
- Short term relief → chronic obstruction
- Management
 - ⊙ Prevention
 - ⊙ Intranasal steroid
 - ⊙ Surgery e.g. turbinate reduction/turbinectomy

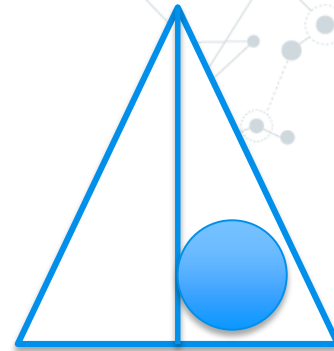


Neoplasm

◎ Rare

◎ Red Flag

- Unilateral obstruction
- Epistaxis
- Bleeding
- Cacosmia
- Proptosis, diplopia, epiphora
- Neurological symptoms

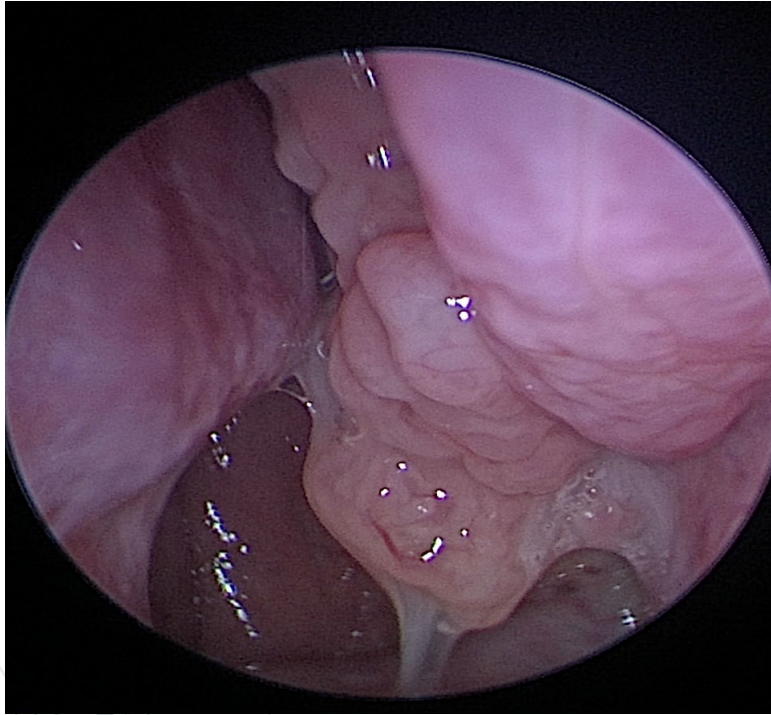


Sinonasal Tumour

Tissue type	Benign	Malignant
Epithelial	Inverted papilloma	Carcinoma Malignant melanoma
Mesenchymal	Juvenile nasopharyngeal angiofibroma	sarcoma
Neural	Meningioma	Olfactory neuroblastoma
Lymphoreticular		Non-Hodgkin's lymphoma
Odontogenic	Ameloblastoma	

And many more...

Inverted Papilloma



- ◎ Human papillomavirus
- ◎ Inverted mucosal surface into stroma of papilloma
- ◎ Benign, locally aggressive
- ◎ Unilateral
- ◎ Lateral nasal wall
- ◎ 2-10% risk of **MALIGNANT** transformation
- ◎ Surgery

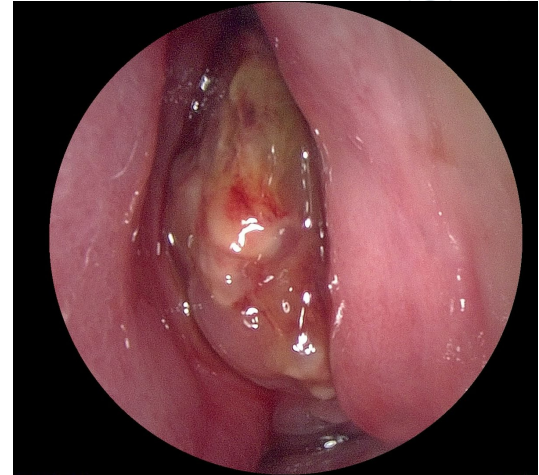
Carcinoma

◎ Risk factors

- smoking
- hard-wood exposure for adenocarcinoma

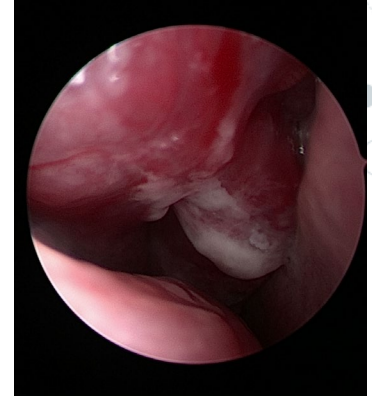
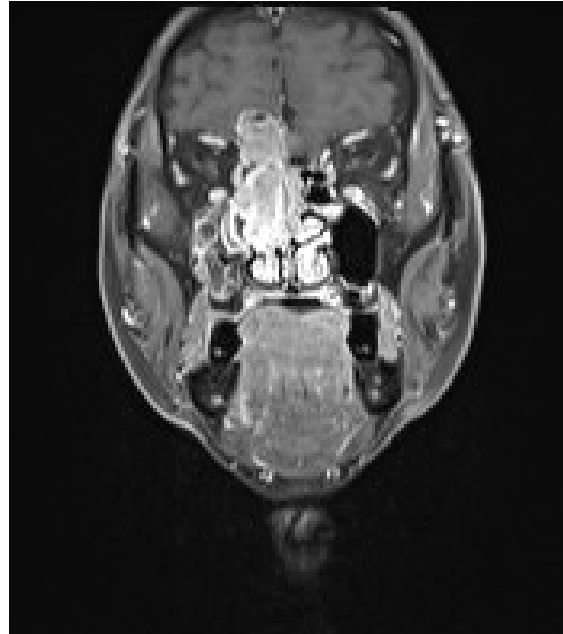
◎ Types

- Squamous cell carcinoma
- Adenocarcinoma
- Sinonasal undifferentiated (anaplastic) carcinoma
- Adenoid cystic carcinoma



Olfactory Neuroblastoma

- © Olfactory epithelium
- © Late presentation with intracranial extension
- © Craniofacial/ cranionasal resection with adjuvant radiotherapy



Sinonasal tumour

Investigations:

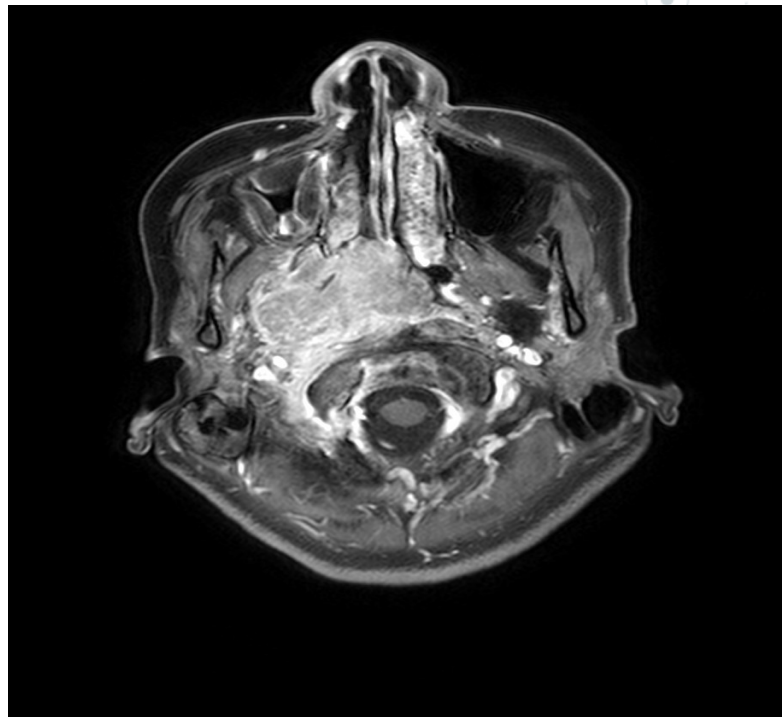
- Computed tomography
- Magnetic Resonance Imaging

Management

- Surgery
 - Endoscopic approach
 - Open approach
 - Craniofacial/ cranionasal resection and reconstruction
- Adjuvant chemotherapy, radiotherapy

Nasopharyngeal Carcinoma

- ◎ >70% new cases in east and southeast Asia
- ◎ 10th most common cancer in Hong Kong¹
- ◎ Global gradual decline in incidence
- ◎ M:F 2.5:1



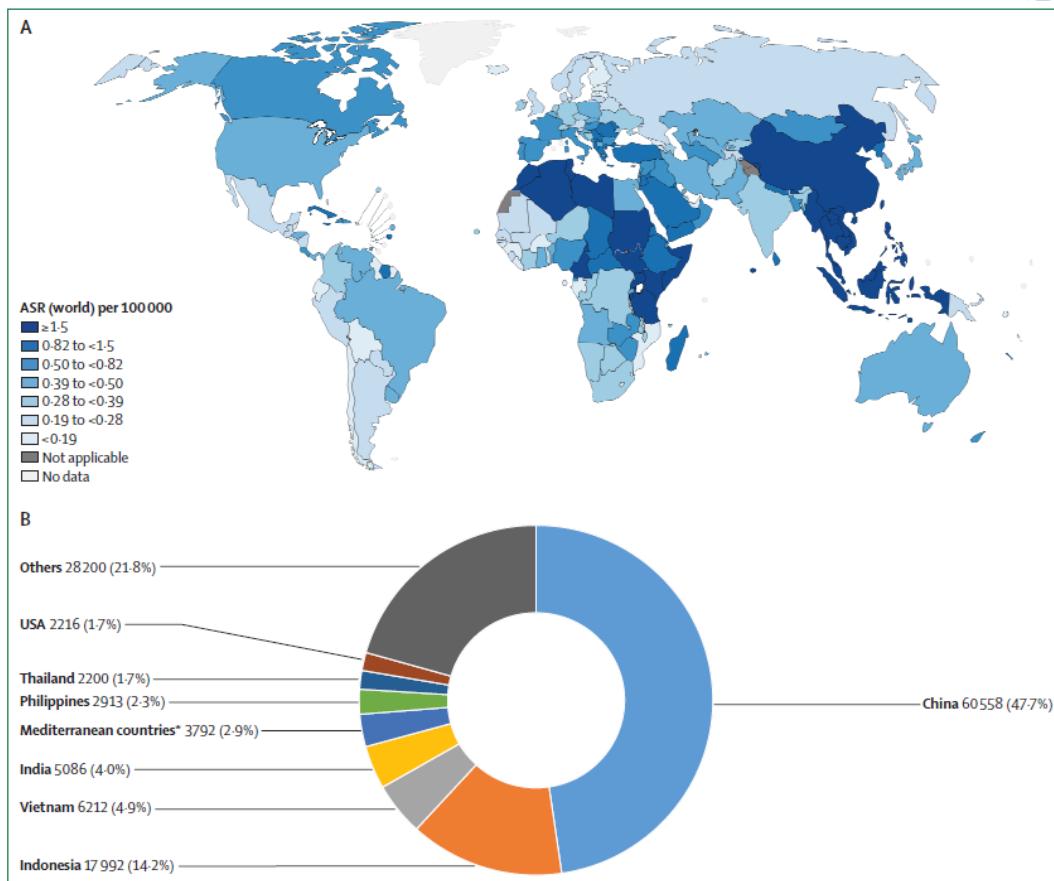


Figure 1: Worldwide distribution of nasopharyngeal carcinoma in 2018

Pathology

◎ World Health Organization

- Non-keratinizing
 - ◎ Differentiated
 - ◎ Undifferentiated
 - ◎ >95% cases in endemic area
 - ◎ Epstein-Barr virus infection
- Keratinising squamous
- Basaloid squamous

Risk factors

- ◎ Epstein-Barr virus infection
- ◎ Host genetics
 - Family history
- ◎ Environmental factors
 - Active and passive tobacco smoking
 - Alcohol
 - Preserved foods
 - Oral hygiene

Clinical Features

- ◎ Common site of origin:
 - Fossa of Rosenmüller
 - ◎ Superior and posterior to the Eustachian tube cartilage
- ◎ Nose: epistaxis, nasal obstruction
- ◎ Ear: hearing loss, tinnitus, aural fullness
- ◎ Eye: diplopia
- ◎ Head and Neck: facial numbness, neck mass

Population Screening

◎ Anti-EBV IgA antibodies

- Early antigen (EA-IgA)
- Viral capsid antigen (VCA-IgA)
- Nuclear antigen 1 (EBNA1-IgA)

Low sensitivity
Low specificity
for screening in
asymptomatic
participants

Population Screening

- ◎ Serum EBV DNA

- At risk group

- ◎ Eg positive family history with first degree relatives

Investigations

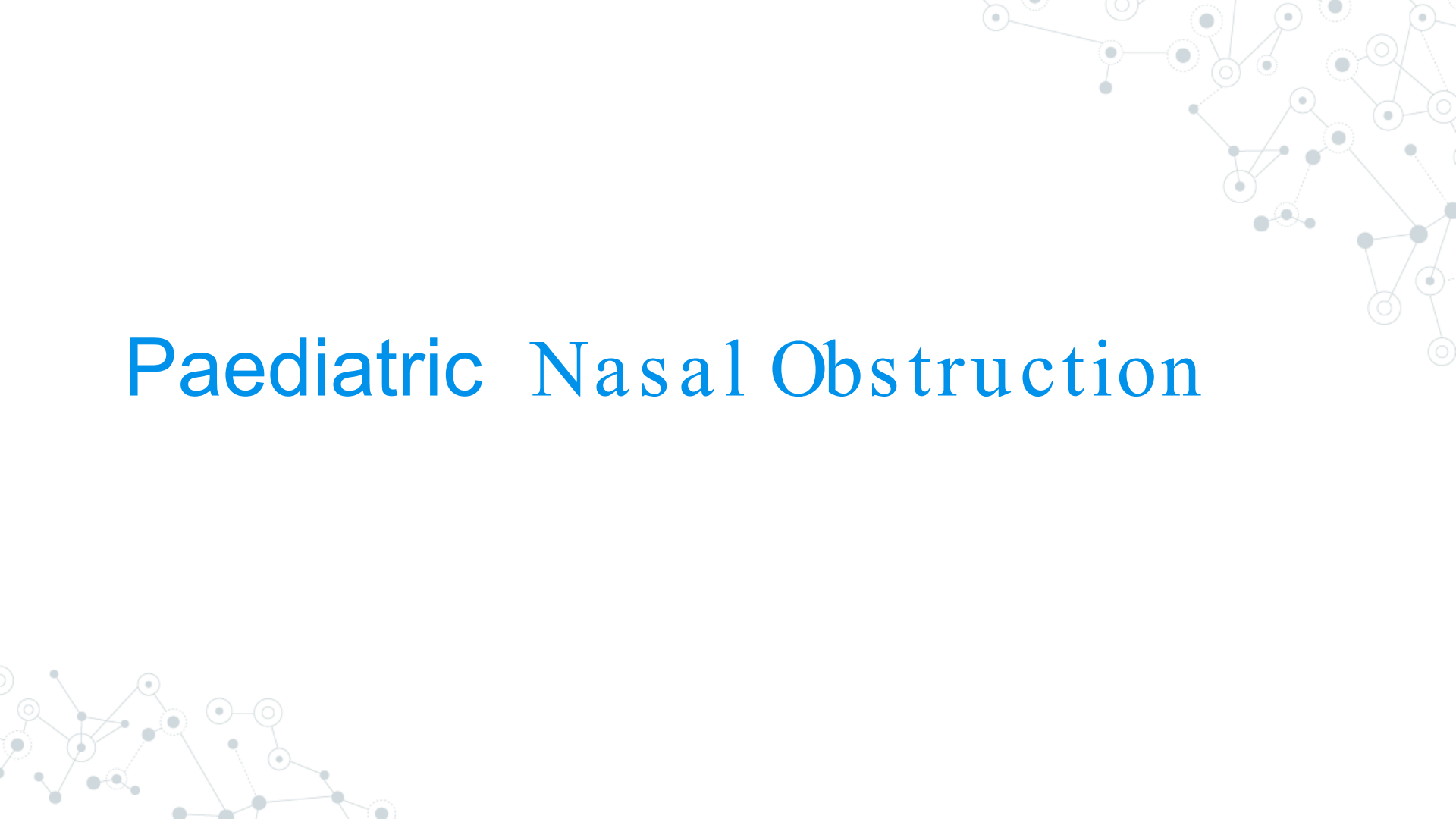
- ◎ Nasoendoscopy + biopsy
- ◎ Ultrasound neck +/- fine needle aspiration cytology
- ◎ Magnetic resonance imaging with contrast
- ◎ ^{18}F -FDG-PET/CT

Nasopharyngeal Carcinoma

◎ Treatment

- Early stage: intensity-modulated radiotherapy (IMRT)
- Late stage: concurrent chemotherapy and radiotherapy
- Residual disease/ recurrence:
 - ◎ Surgery eg open maxillary swing, endoscopic or robotic nasopharyngectomy
 - ◎ *Chemotherapy, second dose radiotherapy*
 - ◎ *Immunotherapy*

Paediatric Nasal Obstruction



Paediatric Nasal Obstruction

- ◎ Neonates
 - Choanal atresia
- ◎ Infant
 - Encephalocele
- ◎ Toddler
 - Adenoid hypertrophy
- ◎ Children
 - Allergic rhinitis
- ◎ Ciliary dysfunction, cystic fibrosis
- ◎ Foreign body

Choanal Atresia

- ◎ Obligate nasal breather
- ◎ Cyclical hypoxia
 - Hypoxia → cry → relief → close mouth → hypoxia
- ◎ Growth failure
- ◎ Urgent treatment for bilateral cases

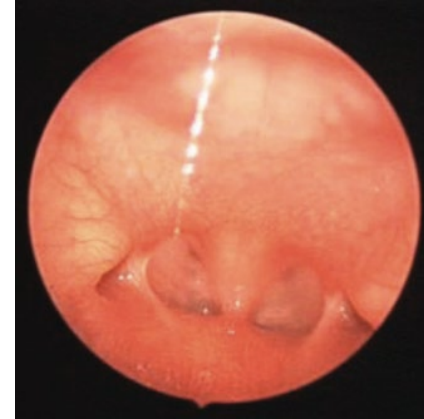
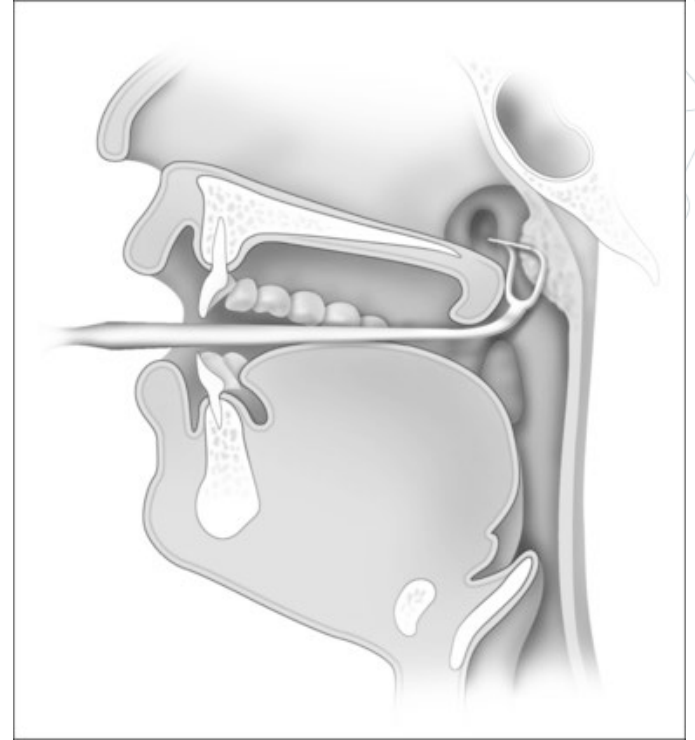


Image source: Logan Turner Disease of the Nose, Throat and Ear- Head and Neck Surgery

Adenoid Hypertrophy

- ◎ Produce B-cells
- ◎ Involute in later childhood, largely disappear in early adulthood
- ◎ Indications for adenoidectomy:
 - Obstructive sleep apnoea syndrome
 - Recurrent rhinosinusitis
 - Recurrent otitis media with effusion



Foreign Body in Nose

◎ Clinical features:

- Irritable child
- Unilateral foul-smelling nasal discharge, sometimes blood-stained
- Excoriation around the nostril
- Occasionally radio-opaque

◎ Risks:

- Button battery → septal perforation
(EMERGENCY)
- Local spread of infection → sinusitis/ meningitis
- Inhalation of foreign body → aspiration
- Injury from clumsy attempts at removal by unskilled person

Take Home Message

- ◎ Sinonasal tumours are rare but present with common nasal symptoms
- ◎ Nasoendoscopy, computed tomography and magnetic resonance imaging are invaluable tools to investigate patients with nasal diseases

Recommended Resources

- ◎ Lecture Notes- Diseases of the Ear, Nose and Throat
- ◎ Ear, Nose and Throat and Head and Neck Surgery. An illustrated colour text.

Reference

- ◎ Logan Turner's Diseases of the Nose, Throat and Ear: Head and Neck Surgery, 11th Edition
- ◎ Allergic Rhinitis and its Impact on Asthma (ARIA) guidelines